

**NATIONAL UNIVERSITY OF MODERN
LANGUAGE ISLAMABAD**

Lab 12



Submitted to:

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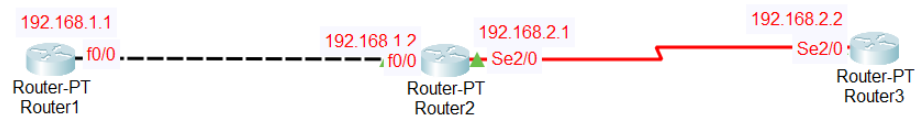
Subject: computer network

Department: BSSE-5-Afternoon

Date: 13-12-2024

Objectives

- Configure the hostname of a router.
- Configure the password for a router.
- We will configure three Cisco routers.
- We will enable these routers for dynamic routing using the RIP protocol



Lab tasks

Task1:

```
Router(config)#
Router(config)#hostname zulqarnain-sp21379
zulqarnain-sp21379(config)#enable password 21379
zulqarnain-sp21379(config)#enable secret 21379
The enable secret you have chosen is the same as your enable password.
This is not recommended. Re-enter the enable secret.
zulqarnain-sp21379(config)#
zulqarnain-sp21379(config)#
```

Task2:

```
zulqarnain-sp21379(config)#
zulqarnain-sp21379(config)#line con 0
zulqarnain-sp21379(config-line)#password 21379
zulqarnain-sp21379(config-line)#login
zulqarnain-sp21379(config-line)#
zulqarnain-sp21379(config-line)#
```

Task3:

```
zulqarnain-sp21379(config)#
zulqarnain-sp21379(config)#line aux 0
zulqarnain-sp21379(config-line)#password 21379
zulqarnain-sp21379(config-line)#login
zulqarnain-sp21379(config-line)#
zulqarnain-sp21379(config-line)#
```

Task4: host name.

Router1:

```
zulqarnain-sp21379(config)#hostname Islamabad
Islamabad(config)#
Islamabad(config)#
Islamabad(config)#
```

Router2:

```
Router(config)#
Router(config)#
Router(config)#hostname Lahore
Lahore(config)#
Lahore(config)#
Lahore(config)#
```

C

Router3:

```
Router(config)#
Router(config)#
Router(config)#hos
Router(config)#hostname Karachi
Karachi(config)#
Karachi(config)#
```

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Task 5: IP address to router interfaces

Router1:

```
Islamabad(config)#
Islamabad(config)#in
Islamabad(config)#interface fa0/0
Islamabad(config-if)#ip add 192.168.1.1 255.255.255.0
Islamabad(config-if)#no sh
Islamabad(config-if)#
Islamabad(config-if)#
Islamabad(config-if)#
Islamabad(config-if)#
```

Router2:

```
Lahore(config)#inter
Lahore(config)#interface fa0/0
Lahore(config-if)#ip add 192.168.1.2 255.255.255.0
Lahore(config-if)#no sh
Lahore(config-if)#
Lahore(config-if)#exit
Lahore(config)#inter
Lahore(config)#interface se2/0
Lahore(config-if)#ip add 192.168.2.1 255.255.255.0
Lahore(config-if)#no sh
Lahore(config-if)#clock rate 56000
Lahore(config-if)#
Lahore(config-if)#
```

Router3:

```
Karachi(config)#inter
Karachi(config)#interface se2/0
Karachi(config-if)#ip add 192.168.2.2 255.255.255.0
Karachi(config-if)#no sh
Karachi(config-if)#
```

Task 6:

Router1:

```
Islamabad(config)#
Islamabad(config)#router rip
Islamabad(config-router)#ver 1
Islamabad(config-router)#network 192.168.1.0
Islamabad(config-router)#no auto
Islamabad(config-router)#no auto
Islamabad(config-router)#no auto-summary
Islamabad(config-router)#
```

Router2:

```
Lahore(config)#
Lahore(config)#router rip
Lahore(config-router)#ver 1
Lahore(config-router)#network 192.168.1.0
Lahore(config-router)#network 192.168.2.0
Lahore(config-router)#no su
Lahore(config-router)#no auto
Lahore(config-router)#no auto-summary
Lahore(config-router)#
Lahore(config-router)#
Lahore(config-router)#
```

Router3:

```
Karachi(config)#
Karachi(config)#
Karachi(config)#router rip
Karachi(config-router)#ver 1
Karachi(config-router)#network 192.168.2.0
Karachi(config-router)#no au
Karachi(config-router)#no auto-summary
Karachi(config-router)#
Karachi(config-router)#
Karachi(config-router)#
```

Task 7:

Router1:

```
Islamabad(config)#
Islamabad(config)#in
Islamabad(config)#interface fa0/0
Islamabad(config-if)#de
Islamabad(config-if)#des
Islamabad(config-if)#description connection to headquarter
Islamabad(config-if)#
Islamabad(config-if)#
Islamabad(config-if)#
```

Router2:

```
Lahore(config)#  
Lahore(config)#interface fa0/0  
Lahore(config-if)#des  
Lahore(config-if)#description connection to branch 1  
Lahore(config-if)#  
Lahore(config-if)#exit  
Lahore(config)#interface se2/0  
Lahore(config-if)#des  
Lahore(config-if)#description connection to branch 2  
Lahore(config-if)#  
Lahore(config-if)#
```

Router3:

```
Karachi(config)#  
Karachi(config)#  
Karachi(config)#interface se2/0  
Karachi(config-if)#des  
Karachi(config-if)#description connection to headquarter  
Karachi(config-if)#  
Karachi(config-if)#  
Karachi(config-if)#
```

Task 8: On all routers set MOTD

```
Lahore(config-line)#exit  
Lahore(config)#  
Lahore(config)#  
Lahore(config)#banner motd /---UNAUTHORIZED ACCESS IN NOT ALLOWED---/  
Lahore(config)#  
Lahore(config)#
```

Change the rip version to 2:

Router1:

```
Islamabad(config-router)#  
Islamabad(config-router)#  
Islamabad(config-router)#ver 2  
Islamabad(config-router)#  
Islamabad(config-router)#  
Islamabad(config-router)#  
Islamabad(config-router)#
```

Router2:

```
Lahore(config)#  
Lahore(config)#router rip  
Lahore(config-router)#  
Lahore(config-router)#ver 2  
Lahore(config-router)#  
Lahore(config-router)#  
Lahore(config-router)#  
Lahore(config-router)#
```

Router3:

```
Karachi(config)#  
Karachi(config)#  
Karachi(config)#router rip  
Karachi(config-router)#ver 2  
Karachi(config-router)#
```

Use debug command to get the details:

```
Islamabad#  
Islamabad#debug ip r  
Islamabad#debug ip ri  
Islamabad#debug ip rip  
RIP protocol debugging is on  
Islamabad#  
Islamabad#  
Islamabad#  
Islamabad#  
Islamabad#  
Islamabad#
```

show ip route

```
Islamabad#  
Islamabad#show ip route  
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP  
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP  
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area  
        * - candidate default, U - per-user static route, o - ODR  
        P - periodic downloaded static route  
  
Gateway of last resort is not set  
  
C    192.168.1.0/24 is directly connected, FastEthernet0/0  
R    192.168.2.0/24 [120/1] via 192.168.1.2, 00:00:22, FastEthernet0/0
```

show ip protocols:

```
Islamabad#  
Islamabad#show ip pro  
Islamabad#show ip protocols  
Routing Protocol is "rip"  
Sending updates every 30 seconds, next due in 12 seconds  
Invalid after 180 seconds, hold down 180, flushed after 240  
Outgoing update filter list for all interfaces is not set  
Incoming update filter list for all interfaces is not set  
Redistributing: rip  
Default version control: send version 2, receive 2  
  Interface          Send Recv Triggered RIP Key-chain  
  FastEthernet0/0    22  
Automatic network summarization is not in effect  
Maximum path: 4  
Routing for Networks:  
  192.168.1.0  
Passive Interface(s):  
Routing Information Sources:  
  Gateway         Distance      Last Update  
  192.168.1.2      120          00:00:10  
Distance: (default is 120)
```